

Discrete Mathematics

CS1111

Lecture 6: Binomial Theorem

Binomial Coefficients

- Consider the binomial expansion of $(a+b)^n$

$$(a + b)^0 = 1$$

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

Binomial Coefficients

- The expansion of $(a + b)^n$ is called the binomial expansion
- The coefficients of the binomial expansion of $(a + b)^n$ are called the binomial coefficients
- One can notice that the binomial coefficients are equal to $C(n, k)$ where $0 \leq k \leq n$, k is the term number
- For example $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = C(3,0)*a^3 + C(3,1)* a^2b + C(3,2)* ab^2 + C(3,3)* b^3$

Binomial Coefficients

Examples

- $(a+b)^2 = a^2 + 2ab + b^2 = C(2,0)*a^2 + C(2,1)*ab + C(2,2)*b^2$
- $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 = C(4,0)*a^4 + C(4,1)*a^3b + C(4,2)*a^2b^2 + C(4,3)*ab^3 + C(4,4)*b^4$
- $(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5 = C(5,0)*a^5 + C(5,1)*a^4b + C(5,2)*a^3b^2 + C(5,3)*a^2b^3 + C(5,4)*ab^4 + C(5,5)*b^5$

Binomial Coefficients

$$C(0,0)$$

$$C(1,0) \quad C(1,1)$$

$$C(2,0) \quad C(2,1) \quad C(2,2)$$

$$C(3,0) \quad C(3,1) \quad C(3,2) \quad C(3,3)$$

$$C(4,0) \quad C(4,1) \quad C(4,2) \quad C(4,3) \quad C(4,4)$$

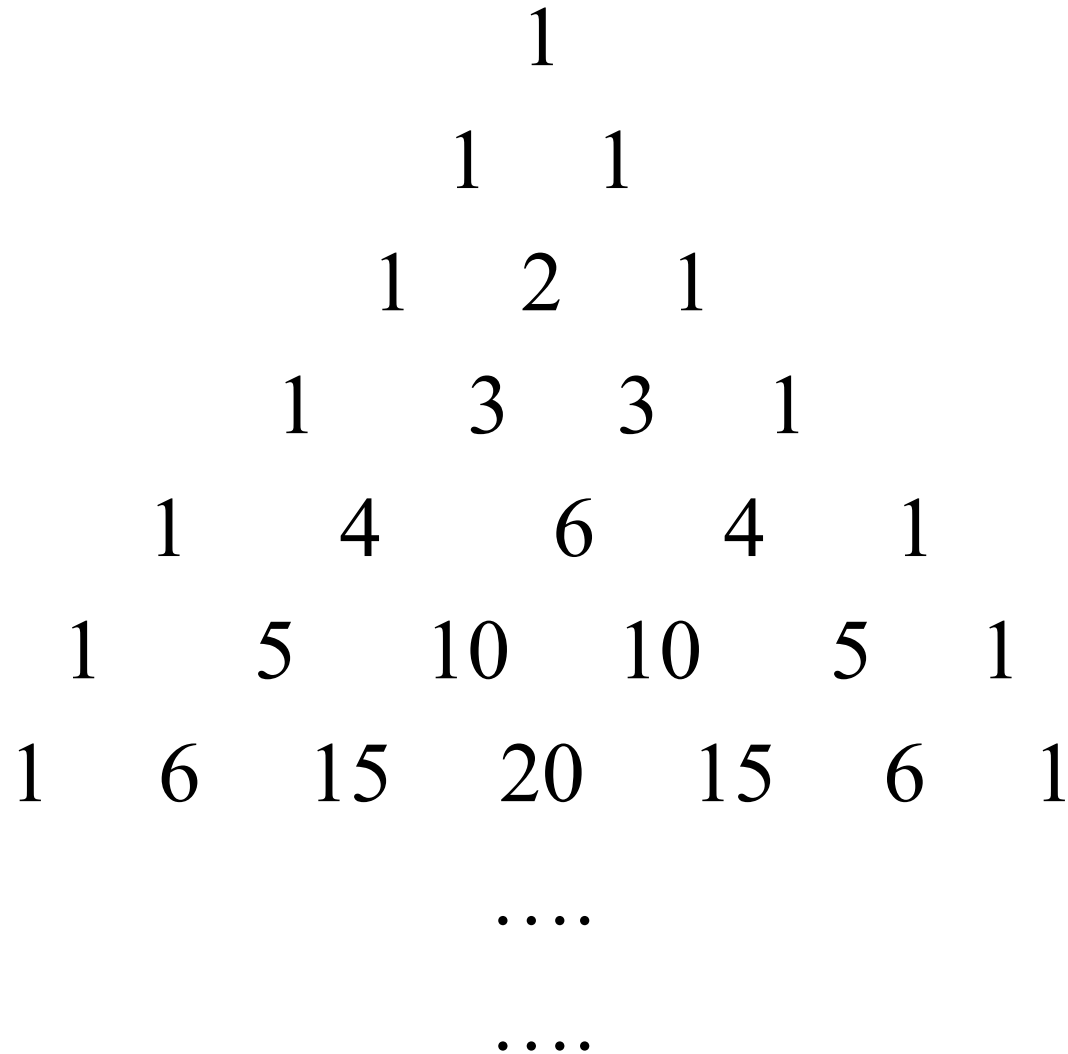
$$C(5,0) \quad C(5,1) \quad C(5,2) \quad C(5,3) \quad C(5,4) \quad C(5,5)$$

$$C(6,0) \quad C(6,1) \quad C(6,2) \quad C(6,3) \quad C(6,4) \quad C(6,5) \quad C(6,6)$$

...

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Pascal Triangle



Binomial Expansion

- Generally, for $n \geq 0$ $(a+b)^n = C(n,0)*a^n b^0 + C(n,1)*a^{n-1}b^1 + C(n,2)*a^{n-2}b^2 + \dots \dots + C(n,k)*a^{n-k}b^k + \dots + C(n,n)*a^0 b^n$
- or

$$(a + b)^n = \sum_{k=0}^n C(n, k) a^{n-k} b^k$$

Pascal Identity

- Let n and k both > 0 and $n \geq k$

$$C(n+1, k) = C(n, k-1) + C(n, k)$$

$$\begin{aligned} \text{RHS: } C(n, k-1) + C(n, k) &= n! / ((n-k+1)!(k-1)!) + \\ & n! / ((n-k)! k!) = n! / ((n-k+1)(n-k)!(k-1)!) + n! / \\ & ((n-k)! k(k-1)!) = n!(k + (n - k + 1)) / (k(n-k+1)(n- \\ & k)!(k-1)!) = n!(n + 1) / (k(n-k+1)(n-k)!(k-1)!) \end{aligned}$$

$$\begin{aligned} &= (n + 1)! / (k(n-k+1)(n-k)!(k-1)!) = (n + 1)! / ((n- \\ & k+1)(n-k)!k(k-1)!) = (n + 1)! / ((n-k+1)(n-k)!k!) = \\ & (n + 1)! / ((n-k+1)!k!) = (n + 1)! / (((n + 1) - k)!k!) = \\ & C(n+1, k) = \text{LHS} \end{aligned}$$

Examples

- $(a + b)^2 = C(2,0) a^2 b^0 + C(2,1) a^1 b^1 + C(2,2) a^0 b^2$
 $= a^2 b^0 + 2a^1 b^1 + a^0 b^2 = a^2 + 2a b + b^2 =$
- $(a + b)^3 = C(3,0)a^3b^0 + C(3,1)a^2b^1 + C(3,2)a^1b^2 +$
 $C(3,3)a^0b^3 = a^3b^0 + 3a^2b^1 + 3a^1b^2 + a^0b^3 = a^3 +$
 $3a^2b + 3ab^2 + b^3$
- $(a + b)^{10} = \sum_{k=0}^{10} C(10,k) a^{10-k} b^k$

Example

- Show that $\sum_{k=0}^n C_k^n = 2^n$
- Using Binomial theorem

$$2^n = (1 + 1)^n = \sum_{k=0}^n C_k^n \cdot 1^{n-k} 1^k = \sum_{k=0}^n C_k^n$$

$$\begin{aligned} 2^n &= (1 + 1)^n = \\ &= \binom{n}{0} \cdot 1^n + \binom{n}{1} \cdot 1^{n-1} \cdot 1 + \binom{n}{2} \cdot 1^{n-2} \cdot 1^2 + \cdots + \binom{n}{n-1} \cdot 1 \cdot 1^{n-1} + \binom{n}{n} \cdot 1^n \\ &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} \end{aligned}$$

Vandermonde Identity

- Let m, n and r be nonnegative integers, $r \leq m, n$, then

$$C(m+n, r) = \sum_{k=0}^r C(m, r-k)C(n, k)$$

- If there are two sets one with m elements and the other with n elements then there are $C(m+n, r)$ ways to select r elements from the $m+n$ collection of elements (LHS)
- For $0 \leq k \leq r$ there are $C(m, r-k)$ ways to select $r-k$ elements from the first set and $C(n, k)$ to select k elements from the second set

Vandermonde Identity

- To select k elements from the second set and $r-k$ elements from the first set there are $C(m, r-k) \times C(n, k)$ ways
- Hence the total number of ways to select r elements from the collection of the two sets is

$$\sum_{k=0}^r C(m, r-k)C(n, k)$$

- therefore

$$C(m+n, r) = \sum_{k=0}^r C(m, r-k)C(n, k)$$

Example

- Using binomial theorem find 15^4

$$\begin{aligned} 15^4 &= (10 + 5)^4 = C(4,0) 10^4 5^0 + C(4,1) 10^3 5^1 + \\ &\quad C(4,2) 10^2 5^2 + C(4,3) 10^1 5^3 + C(4,4) 10^0 5^4 = \\ &\quad 10^4 \times 5^0 + 4 \times 10^3 \times 5^1 + 6 \times 10^2 \times 5^2 + 4 \times 10^1 \times 5^3 + \\ &\quad 10^0 \times 5^4 = 10^4 + 4 \times 10^3 \times 5 + 6 \times 10^2 \times 5^2 + 4 \times 10 \times 5^3 + \\ &\quad 5^4 = 10000 + 4 \times 1000 \times 5 + 6 \times 100 \times 25 + 4 \times 10 \times 125 \\ &\quad + 625 = 10000 + 20000 + 15000 + 5000 + 625 = \\ &\quad 30000 + 20000 + 625 = 50625 \end{aligned}$$